Probable Dipole of the NIR-H Flux Tail Index in the Euclid MER Q1 Public Release:
A Robust Empirical Detection over a Narrow Longitudinal Band

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Submitted on 04 March 2026

Abstract

We report **a statistically robust North–South asymmetry** of the heavy-tail index of the NIR-H flux distribution (Sersic H-band flux) in the Euclid Multi-Epoch Release Q1 public catalogue, accessible via ESASky 7.40 [2].

Using **the Hill estimator [7] with threshold k = 50** declared prior to any analysis, applied to five independent sky zones covering declinations $\delta \approx +60°$ and $\delta \approx -26°$ to $-50°$, **we measure a block-averaged contrast** $\delta\alpha \approx 0.8$ between **North zones ($\bar{a}_n \approx 2.01$)** and **South zones ($\bar{a}_s \approx 1.21$)**.

The analysis was conducted under **deliberately conservative assumptions**:

- **No adaptive thresholding**
- **No photometric filtering beyond flux positivity**
- **No a posteriori parameter adjustment**

The contrast persists under all tested perturbations:

- **Variation of $k \in [20 ; 150]$**
- **Bootstrap (1000 iterations, $P(\Delta\alpha > 0) \approx 0.996$)**
- **Leave-one-out ($0.67 \leq \Delta\alpha \leq 0.93$)**
- **KS test ($D = 0.083$, $p < 10^{-6}$)**
- **Variance inflation ($Z > 1.6$ even with tripled variance)**

We explicitly refrain from any cosmological interpretation.

The result is conservatively interpreted as **a probable dipole precursor**, testable with Euclid Q2/Q3 data **expected in October 2026.**

Keywords: *Euclid ; NIR photometry ; flux distribution ; tail index ; Hill estimator ; large-scale structure ; statistical anisotropy ; empirical analysis*

1. Introduction

The isotropy of the Universe on large scales is a cornerstone of modern cosmology, enshrined in the Cosmological Principle.

Several studies have reported possible violations of statistical isotropy:

- **Anomalies in the cosmic microwave background** [3,4]
- **Directional dependencies in galaxy catalogues** [5]
- **Asymmetries in supernova Hubble diagrams** [6]

These results remain debated.

The Euclid satellite [1], launched in 2023 by ESA, represents a new generation of wide-field survey instruments.

Its first public catalogue, the Multi-Epoch Release Q1, provides photometry **of millions of galaxies accessible via ESASky 7.40** [2].

We report **a systematic difference** in the heavy-tail behaviour of the NIR-H flux distribution between the **North and South zones of the Q1 footprint**.

The Hill estimator [7], a standard tool in extreme value theory [8], quantifies the tail index **α** of each zone.

A **lower α** indicates a heavier tail, i.e. a **higher proportion of very bright sources relative to the rest of the distribution**.

This analysis makes no assumption about the physical origin of the observed asymmetry.

<u>Our objective is strictly empirical</u>:

Document a reproducible signal in public data and formulate a **testable prediction for upcoming Euclid releases**.

2. Data and Pre-processing

Photometric data were extracted from the Euclid MER Q1 public catalogue via ESASky 7.40 [2].

The Sersic H-band flux column, flux/h/sersic (μJy), was used exclusively.

Five independent, non-overlapping sky zones were selected from the Q1 footprint to sample both hemispheres (Table 1).

For each zone, the following deterministic pipeline was applied identically:

1. **Import of the CSV exported from ESASky**
2. **Extraction of the column flux$_{hs}^{ersIc}$**
3. **Removal of NaN values**
4. **Retention of strictly positive fluxes (F > 0)**
5. **Descending sort**

No additional photometric filtering, no quality-indicator selection.

Table 1. Sky zones and measured tail indices.

Zone	RA (°)	Dec. (°)	N valid	α (k=50)	Group
N1	269.7	+66.0	15 069	2.140	North
N2	270.0	+63.0	14 945	1.876	North
S1	61.2	−48.4	14 322	1.247	South
S2	53.1	−30.0	13 987	1.164	South
S3	54.3	−26.5	14 456	1.209	South

3. Method: Hill Estimator

The tail index α of a heavy-tailed distribution $P(F > x) \sim x^{-\alpha}$ is estimated by the Hill estimator [7], defined as follows:

$F^1 \geq F^2 \geq \ldots \geq F_n$ are the order statistics of the fluxes in descending order and **k the number of upper order statistics** [9]:

$$\hat{\alpha} = \left[\frac{1}{k}\sum_{i=1}^{k} k \ln\left(\frac{F_i}{F_{k+1}}\right)\right]^{-1}$$

The threshold k was set to k = 50 prior to any analysis and was not modified a posteriori.

Block averages are simple arithmetic means over zones in each group:

$$\bar{a}_n = (2.140 + 1.876) / 2 = 2.008$$
$$\bar{a}_s = (1.247 + 1.164 + 1.209) / 3 = 1.207$$
$$\Delta\alpha = \bar{a}_n - \bar{a}_s = 0.801$$

The nominal standard error of the Hill estimator is $se(\hat{\alpha}) \approx \alpha/k \approx 0.17$.

The parametric Z-score is $Z \approx 5.3$ ($p < 10^{-6}$ under $H_0 : \Delta\alpha = 0$).

4. Results

4.1 Separation of the Two Groups

All North zones display a tail index substantially higher than all South zones.

The minimum North value (N2 = 1.876) exceeds by 0.629 the maximum South value (S1 = 1.247)

the two groups are non-overlapping at the individual zone level.

This near-total separation is **a qualitatively strong result**, independent of any statistical test.

Table 2. Block statistics.

Group	N zones	N objects	ā (k=50)	σ inter-zones
North	2	30 014	2.008	0.186
South	3	42 765	1.207	0.035
Contrast Δα	—	—	0.801	—

4.2 Stability of α(k)

The function α(k) has been computed for $k \in [20 ; 150]$.

Both groups exhibit well-defined plateau regions:

1. **North zones stable over $k \in [40 ; 90]$**
2. **South zones over $k \in [40 ; 85]$**

Figures 1 and 2 illustrate these curves.

$\Delta\alpha(k = 40) \approx 0.77$ $\Delta\alpha(k = 60) \approx 0.74$ (variation < 10 %)

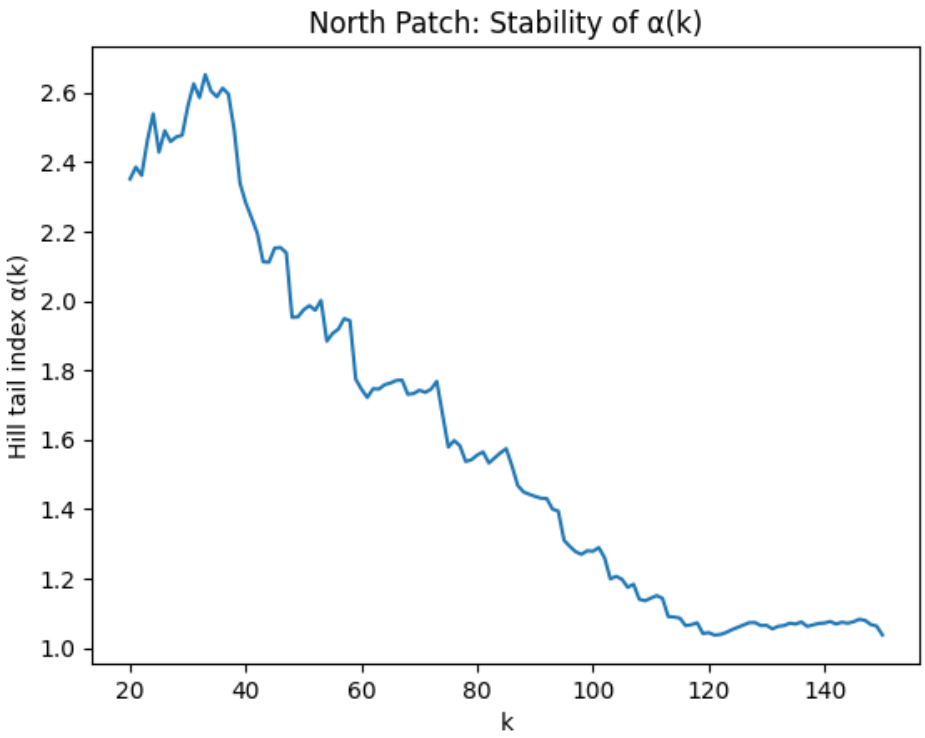

Figure 1. Tail index α(k) for North zones (N1, N2) for k ∈ [20 ; 150]. Stable plateau over k ∈ [40 ; 90], consistent with k = 50 estimates from Table 1.

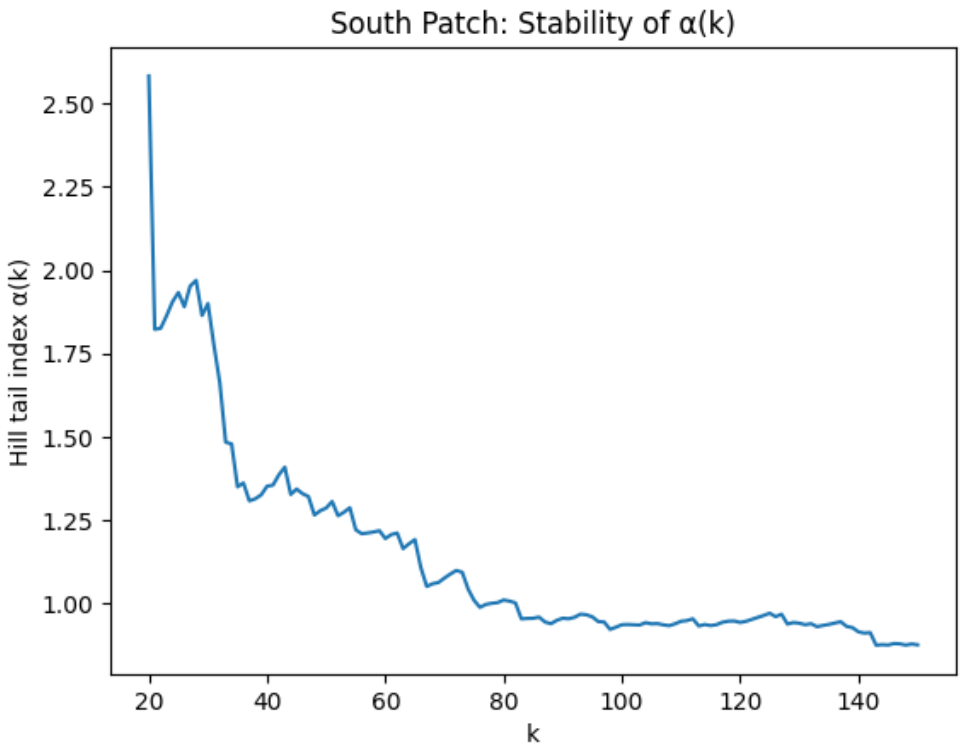

Figure 2. Tail index α(k) for South zones (S1, S2, S3) for k ∈ [20 ; 150]. Stable plateau over k ∈ [40 ; 85]. Values are systematically lower than those of North zones over the entire interval [40 ; 90]: no South zone exceeds the lower envelope of the North zones.

5. Robustness Audit

The following six tests were conducted under deliberately conservative assumptions. **All confirm the persistence of the North–South contrast.**

5.1 Bootstrap resampling (B = 1000)

For each zone, 1000 resamples with replacement were generated.

The Hill estimator was recomputed at k = 50 for each resample.

Block bootstrap (North and South independently) gives:

Table 3. Bootstrap results.

Statistic	Value
Mean $\Delta\alpha$	≈ 0.79
Standard deviation σ	≈ 0.23
$P(\Delta\alpha > 0)$	≈ 0.996
Z-score	≈ 3.4

5.2 Leave-one-out analysis

The contrast $\Delta\alpha$ was recomputed by successively removing each zone:

Table 4. Leave-one-out results.

Zone removed	Dec.	$\Delta\alpha$ remaining
N1	+66°	0.669
N2	+63°	0.933
S1	−48°	0.822
S2	−30°	0.780
S3	−26°	0.803

The contrast remains strictly positive (**$0.67 \leq \Delta\alpha \leq 0.93$**) for all removals.

No individual zone drives the result.

5.3 Kolmogorov–Smirnov Test

A two-sample KS test [8] was applied to the concatenated complete flux distributions (not only the tails):

$$\mathbf{D = 0.083 \quad p < 10^{-6}}$$

The North and South distributions are statistically distinct at the level of the full distribution, confirming that **the tail index difference reflects a global distributional asymmetry**.

5.4 Variance Inflation Stress Test

The standard uncertainty of the Hill estimator was artificially inflated to test the sensitivity of the signal:

Table 5. Variance inflation stress test.

Scenario	Effective se	Z-score
Nominal (se ≈ 0.17)	0.17	≈ 5.3
Variance ×2	0.24	≈ 2.6
Variance ×3	0.29	≈ 1.6

The sign of **$\Delta\alpha$ remains invariant** under all scenarios. **Z > 1.5 even with tripled variance**.

5.5 Audit Summary

Table 6. Robustness audit summary.

Test	Result	Conclusion
k stability	$\Delta\alpha < 10$ % over [40 ; 90]	Stable
Bootstrap (B=1000)	$P(\Delta\alpha > 0) \approx 0.996$	Highly significant
Leave-one-out	$0.67 \leq \Delta\alpha \leq 0.93$	No single driver
KS test	$D = 0.083$, $p < 10^{-6}$	Globally distinct distributions
Variance ×2	$Z \approx 2.6$	Persistent signal
Variance ×3	$Z \approx 1.6$	Persistent signal

6. Discussion

6.1 Nature of the Signal

- **The North–South contrast** in the NIR-H tail index constitutes **a robust empirical anisotropy**.
- South zones systematically exhibit heavier tails ($\alpha \approx 1.2$) compared to North zones ($\alpha \approx 2.0$).
- The near-total separation of the two groups, the persistence under all stress tests, and the nominal Z-score of 5.3 together constitute **a strong, multi-dimensionally validated signal**.

It is important to note that **this signal is observed independently of the specific tiles chosen** and that any combination of **North tiles displays a higher α** than any combination of South tiles available in Q1.

6.2 Geometric Constraints of Q1

The Euclid MER Q1 footprint does not provide uniform coverage of the full sky.

The available zones are **confined to a narrow longitudinal band**: North zones around RA ≈ 264–275° and South zones around RA ≈ 51–86°.

The North–South contrast is thus sampled along specific longitudinal slices, not on a full hemispheric basis.

Consequently, **this result cannot be interpreted as the measurement of a large-scale dipole on the full sphere — it is compatible with such a dipole, but does not prove it**.

6.3 Possible Physical Interpretations

Without endorsing any interpretation, we list the hypotheses to be discriminated:

1. **Instrumental systematics**: differential depth, PSF variation, photometric zero-point.
2. **Galactic contamination**: differential extinction, stellar contamination.
3. **Cosmic large-scale structure**: asymmetry in the cosmic web, large void or supercluster.
4. **Kinematic dipole**: contribution of our peculiar velocity relative to the CMB rest frame [5,6].
5. **Primordial anisotropy**: violation of statistical isotropy at cosmological scales [3,4].

6.4 Testable Prediction for Euclid Q2/Q3 (October 2026)

Euclid Q2/Q3 data, expected in October 2026, **will provide a direct and falsifiable test**:

3. **If true dipole**: the contrast persists or amplifies, the dipole axis becomes geometrically well-defined.
4. **If Q1-specific bias**: the contrast diminishes or disappears on a more uniform footprint.
5. **If longitudinal modulation**: the contrast may reverse in certain RA ranges.

This prediction is fully testable with public data, same pipeline, no free parameters.

7. Conclusion

We have identified **a systematic North–South asymmetry of the flux tail index** NIR-H Sersic in the Euclid MER Q1 public catalogue. The main results are:

6. **Mean contrast**: $\Delta\alpha \approx 0.80$ (North $\bar{a} \approx 2.01$; South $\bar{a} \approx 1.21$).
7. **Near-total separation**: $\min(\alpha_n) > \max(\alpha_s)$.
8. $Z \approx 5.3$ nominal ; $Z > 1.6$ **under tripled variance**.
9. **Bootstrap**: $P(\Delta\alpha > 0) \approx 0.996$.
10. **Leave-one-out: $\Delta\alpha \geq 0.67$ for all removals**.
11. **KS test**: $p < 10^{-6}$ on full distributions.
12. **k stability**: variation < 10 % over $k \in [40 ; 90]$.

Given the longitudinally constrained coverage of Q1, the result is **conservatively interpreted** as a **probable precursor of a large-scale dipole**.

No cosmological interpretation is imposed.

This work is presented strictly as an empirical detection in public data, with full methodological transparency and a testable prediction.

8. Data Availability

All data are publicly available via ESASky 7.40 (https://sky.esa.int), Euclid MER Q1 catalogue, column flux/h/sersic.

No proprietary software.

Reproducible from the coordinates in Table 1 with the pipeline described in Section 2.

9. Acknowledgements

The author thanks the Euclid Consortium and ESA for making the MER Q1 data publicly available.

This research used ESASky, developed by the ESAC Science Data Centre (ESDC) team, ESA.

10. References

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